

PHANIDHAR AKULA

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AI & Full-Stack Software Engineer • Production AI, Distributed Systems & ML • Authorized to work in the U.S. Oct 2026 – Oct 2029 (OPT + STEM OPT)

EDUCATION

Miami University | M.S. in Computer Science

Aug 2024 – Aug 2026

Oxford, OH • GPA: 3.86/4.00 • Full Scholarship • Assistantship

Thesis: SimForge – A Reproducible, Cross-Simulator Benchmarking Framework for Urban Traffic Simulation (committee of three faculty advisors)

Coursework: Generative AI, Machine Learning, Advanced Database Systems, Software Quality Assurance, Cryptography

Malla Reddy University | B.Tech in Computer Science

Aug 2020 – May 2024

Hyderabad, India • CGPA: 8.2/10 • Two peer-reviewed publications (CNN soil classification, LDA topic modeling)

SELECTED PROJECTS

LumiAI | Production AI Education Platform – studywithlumi.com

Nov 2025 – Present

React, TypeScript, Supabase (PostgreSQL/Auth/Storage), LLMs, Vercel serverless, custom domain

- Solo-architected, built, deployed, and operate a production AI tutoring platform serving 55+ active student users, with document-grounded AI chat, auto-generated quizzes and flashcards, SM-2 spaced-repetition review, and voice mode.
- Built the AI layer end to end: LLM integration via a token-verified serverless function (server-side API-key isolation; anonymous calls rejected), document-grounded (RAG-style) context from user-uploaded materials, and streaming responses with prompt caching.
- Owned the full stack: React/TypeScript frontend, PostgreSQL with row-level security, Google OAuth, hardened SECURITY DEFINER database functions, and an admin analytics panel; shipped and iterated on a custom domain.

Cityscape | Open-Source Population Synthesis & Geospatial Demand Generation

Aug 2025 – Present

C++, OpenMP, OSM / Geofabrik, US Census PUMS, OSC HPC (Pitzer)

- Authored three merged C++ pull requests (github.com/raodj/cityscape, PRs #1–#3) to an open-source population-synthesis and trip-demand engine. Fixed building-classification and data-integrity logic across five metros (e.g. LA ~70% → ~5% synthetic ratio), removing ~1.25M erroneous records and affecting well over 1M data points.
- Exposed configurable CLI parameters and ran a 4-city, 16-combination parameter sweep on HPC, raising worst-case calibration accuracy (R^2) from 0.812 to 0.917.

SimForge | Master's Thesis – A Reproducible, Cross-Simulator Benchmarking Framework for Urban Traffic Simulation

Aug 2024 – Present

Python, OpenMP, multiprocessing, GPU-accelerated simulation, SLURM, OSC HPC (Pitzer / Cardinal), SUMO, MATSim, DTALite, OSM, US Census PUMS

- Built a reproducible benchmarking framework integrating three simulation engines under one contract with byte-identical reproducibility (SHA-256-hashed inputs) and automated validation (~570 pytest tests, mutation testing, statistical CIs; 76% coverage).
- Engineered a parallelized routing pipeline that cut runtime from ~68h to ~5–8h (~25–30× speedup) on 500K-scenario workloads via deduplication, 16-way multiprocessing. Pool parallelism, and content-addressed caching across two HPC clusters.
- Scaled to ~80K nodes / ~200K links across Chicago, NYC, and LA, calibrated against US Census microdata (2.4M–6.8M persons); evaluated GPU-accelerated simulators (LPSim, QarSUMO on V100) before scoping to the final engine set.
- Designed a state-aware routing pre-processor (BFS with turn-restriction and connectivity filtering) that forces engines onto identical canonical routes, isolating engine behavior from input differences.

EXPERIENCE

Miami University | Graduate Assistant – Department of CSE

Aug 2025 – May 2026

Oxford, OH • Research group of Dr. DJ Rao

- Graduate research in the simulation / digital-twin group; lead developer on Cityscape (above), with work integrated into the SimForge thesis pipeline. Co-administer the Grand Challenges Scholars Program (GCSP) mentoring undergraduate research.
- Prior: SWE intern (Smart Pathfinders, 2024) and freelance full-stack engineer for three startup clients (2023–2024).

LEADERSHIP

- President, Graduate Students of Color Association (104 members; Certificate of Appreciation, 2026) and elected Board Member, Graduate Student Council, Miami University (Aug 2025 – May 2026) – led recruitment, events, cross-team coordination, and reviewing student petitions.

TECHNICAL SKILLS

Languages: Python (advanced), TypeScript, JavaScript, C++, SQL, Java (basic)

AI/ML: LLMs, RAG-style document grounding, prompt engineering, PyTorch, TensorFlow, CNNs, NLP, model inference

Full-Stack & Cloud: React, Node/serverless (Vercel), REST APIs, GraphQL, PostgreSQL, MySQL, Supabase, AWS, Docker, OAuth, row-level security

Distributed & HPC: OpenMP, multiprocessing, GPU-accelerated simulation, SLURM, distributed HPC, content-addressed caching, Linux, Git

Engineering & Testing: pytest, mutation testing, statistical validation (95% CIs), reproducibility guards, code review

Simulation & Data: agent-based & traffic simulation, digital twins, OSM / Geofabrik, US Census PUMS pipelines

PUBLICATIONS & RECOGNITION

► Akula, P., Stojanovski, S., & Rao, D. M. "CITYSCAPE: A Digital Twin Framework for Temporospatial Modeling of Work-Commute Dynamics and Roadway Infrastructure Demand" *Frontiers* (first author; under review, 2026).

► Two peer-reviewed publications in IJSREM (2023): "Topic Modelling of Web Pages with LDA Methods" (DOI: 10.55041/IJSREM27350) · "Soil Image Classification using Deep Learning" (DOI: 10.55041/IJSREM23138).

► 10+ technical certifications in ML/AI, cloud, and software engineering (AICTE, Coursera, NPTEL, Forage, AWS).